

Data Exploration

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Notes

Please refer to the codebook ([click here](#)) for information on the data, especially on understanding what the variables mean.

Predict TUITION2 and TUITION3 (In-state/Out-of-state Tuition)

Helper

```
VAR_TO_KEEP <- c(
  "CONTROL", "LOCALE", "ACTCM50", "SATVR50", "SATMT50",
  "DVADM01", "DVADM04", "STUFACR", "APPLFEEU", "ENRTOT",
  "RMFRGNCP", "RMINSTTP", "RMOUSTTP", "SLO6", "SLOA",
  "CLASIZUND20", "CLASIZOVE50", "PROPBOR", "AVEDEBT",
  "AGRNT_T", "GBA6RTT", "GBA4RTT", "TRRTTOT", "EFNRALT"
)

prep_data <- function(data, target) {
  data %>%
    filter(!is.na(.data[[target]]), .data[[target]] > 0) %>%
    select(all_of(target), all_of(VAR_TO_KEEP)) %>%
    mutate(across(any_of(c("SLO6", "SLOA")), ~ ifelse(. == "Yes", 1, 0))) %>%
    select(where(is.numeric)) %>%
    na.omit()
}

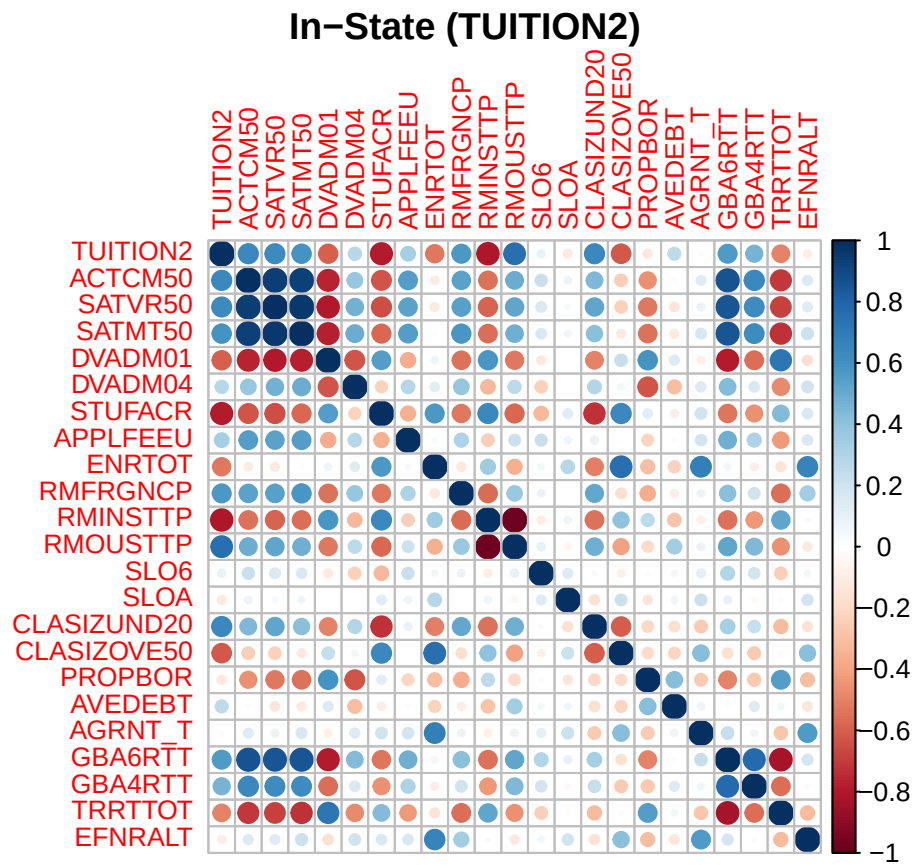
get_best_subsets_modern <- function(df_clean, target_var) {
  form <- as.formula(paste(target_var, "~ ."))
  reg_fit <- regsubsets(form, data = df_clean, nvmax = 10, nbest = 1)
  res_sum <- summary(reg_fit)

  data.frame(
    NumVars = 1:length(res_sum$adjr2),
    AdjR2 = res_sum$adjr2,
    res_sum$which
  ) %>% arrange(desc(AdjR2))
}
```

Corr Plot

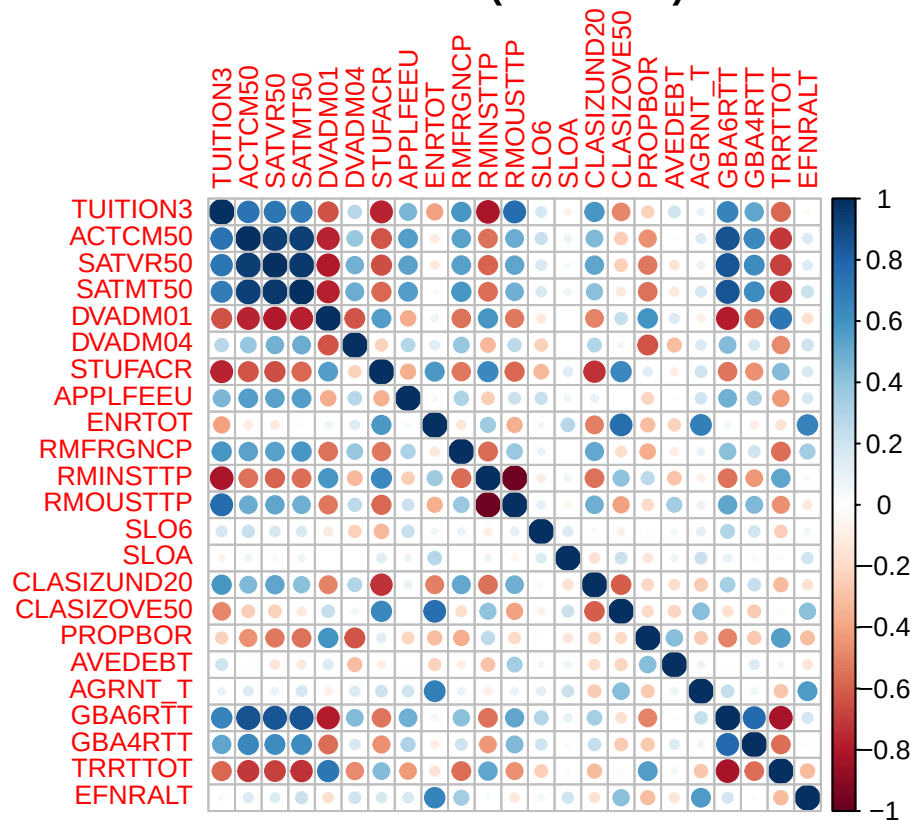
```
# Generate the correlation matrices
df_in  <- prep_data(df, "TUITION2")
df_out <- prep_data(df, "TUITION3")

corrplot(cor(df_in), title = "In-State (TUITION2)", mar = c(0,0,1,0), tl.cex = 0.8)
```



```
corrplot(cor(df_out), title = "Out-of-State (TUITION3)", mar = c(0,0,1,0), tl.cex = 0.8)
```

Out-of-State (TUITION3)

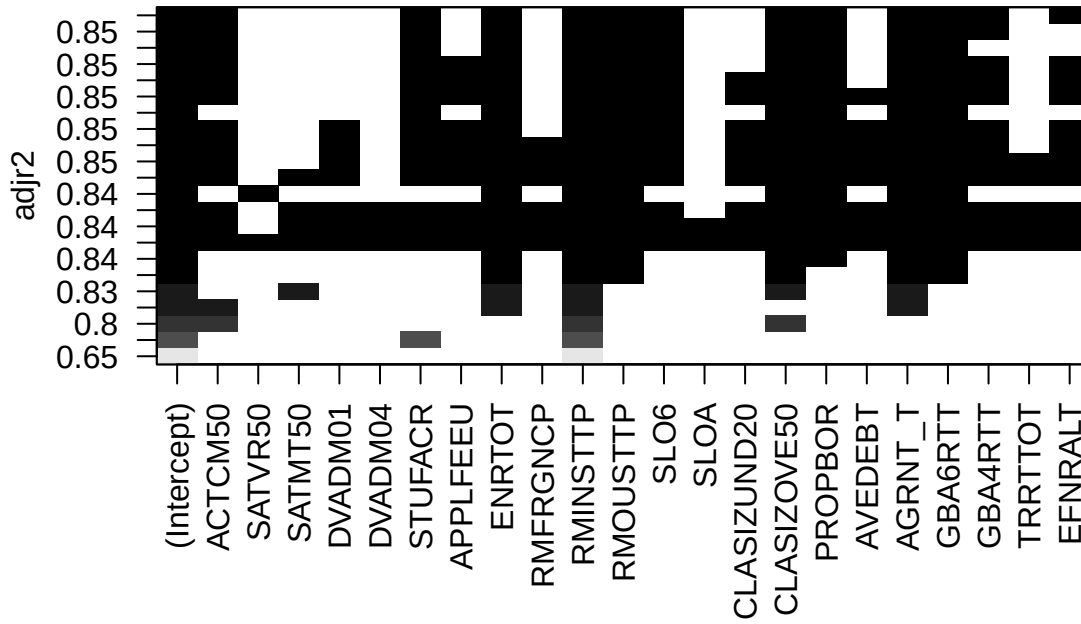


Best Subsets

```
# Run the Analysis
best_in  <- get_best_subsets_modern(df_in, "TUITION2")
best_out <- get_best_subsets_modern(df_out, "TUITION3")

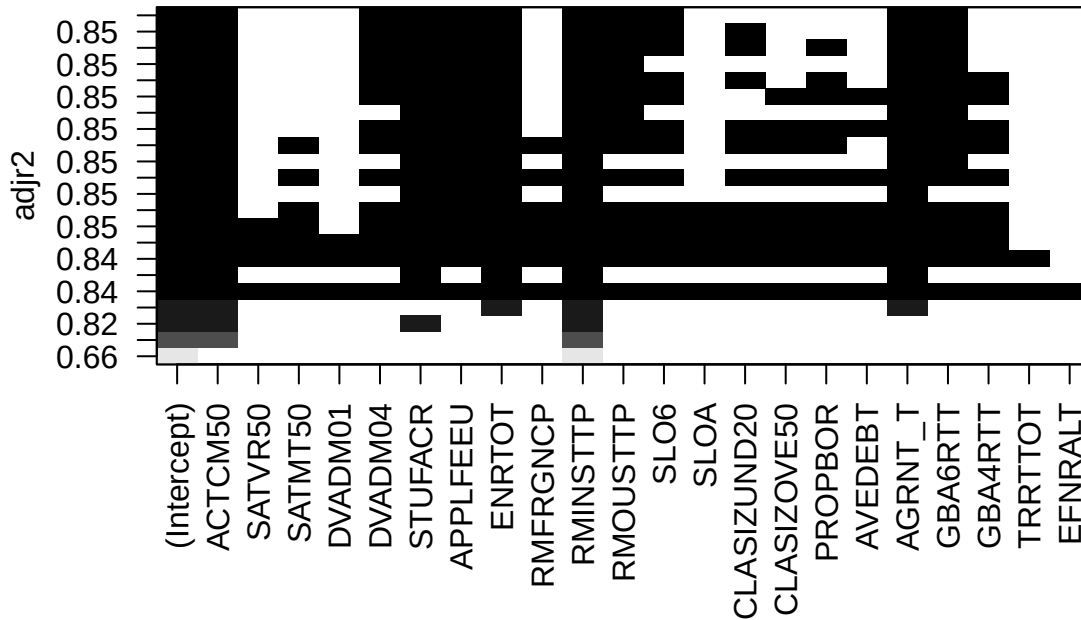
# Visual comparison of models
plot(regsubsets(TUITION2 ~ ., data = df_in, nvmax=24), scale="adjr2", main="In-State Best Subsets")
```

In-State Best Subsets



```
plot(regsubsets(TUITION3 ~ ., data = df_out, nvmax=24), scale="adjr2", main="Out-of-State Best Subsets")
```

Out-of-State Best Subsets



```
# Display numerical results
head(best_in, 5)
```

```
##      NumVars      AdjR2 X.Intercept. ACTCM50 SATVR50 SATMT50 DVADM01 DVADM04
## 10         10 0.8508211          TRUE    TRUE    FALSE    FALSE    FALSE
##  9         9 0.8491440          TRUE    FALSE    FALSE    FALSE    FALSE
##  8         8 0.8447796          TRUE    FALSE    TRUE    FALSE    FALSE
##  7         7 0.8399203          TRUE    FALSE    FALSE    FALSE    FALSE
##  6         6 0.8365765          TRUE    FALSE    FALSE    FALSE    FALSE
##      STUFACR APPLFEEU ENRTOT RMFRGNCP RMINSTTP RMOUSTTP SLO6 SLOA CLASIZUND20
## 10    TRUE    FALSE    TRUE    FALSE    TRUE    TRUE    TRUE    FALSE    FALSE
##  9    TRUE    FALSE    TRUE    FALSE    TRUE    TRUE    TRUE    FALSE    FALSE
##  8    FALSE    FALSE    TRUE    FALSE    TRUE    TRUE    TRUE    FALSE    FALSE
##  7    FALSE    FALSE    TRUE    FALSE    TRUE    TRUE    TRUE    FALSE    FALSE
##  6    FALSE    FALSE    TRUE    FALSE    TRUE    TRUE    TRUE    FALSE    FALSE
##      CLASIZOVE50 PROPBOR AVEDEBT AGRNT_T GBA6RTT GBA4RTT TRRTTOT EFNRLT
## 10         TRUE    TRUE    FALSE    TRUE    TRUE    FALSE    FALSE    FALSE
##  9         TRUE    TRUE    FALSE    TRUE    TRUE    FALSE    FALSE    FALSE
##  8         TRUE    TRUE    FALSE    TRUE    TRUE    FALSE    FALSE    FALSE
##  7         TRUE    TRUE    FALSE    TRUE    TRUE    FALSE    FALSE    FALSE
##  6         TRUE    FALSE    FALSE    TRUE    TRUE    FALSE    FALSE    FALSE
```

```
head(best_out, 5)
```

```
##      NumVars      AdjR2 X.Intercept. ACTCM50 SATVR50 SATMT50 DVADM01 DVADM04
```

```
## 10      10 0.8541484      TRUE      TRUE      FALSE      FALSE      FALSE      TRUE
## 9       9 0.8527058      TRUE      TRUE      FALSE      FALSE      FALSE      TRUE
## 8       8 0.8509133      TRUE      TRUE      FALSE      FALSE      FALSE      FALSE
## 7       7 0.8492592      TRUE      TRUE      FALSE      FALSE      FALSE      FALSE
## 6       6 0.8481619      TRUE      TRUE      FALSE      FALSE      FALSE      FALSE
##      STUFACR APPLFEEU ENRTOT RMFRGNCP RMINSTTP RMOUSTTP SLO6 SLOA CLASIZUND20
## 10      TRUE      TRUE      TRUE      FALSE      TRUE      TRUE      TRUE FALSE      FALSE
## 9      TRUE      TRUE      TRUE      FALSE      TRUE      TRUE      TRUE FALSE FALSE      FALSE
## 8      TRUE      TRUE      TRUE      FALSE      TRUE      TRUE      TRUE FALSE FALSE      FALSE
## 7      TRUE      TRUE      TRUE      FALSE      TRUE      FALSE FALSE FALSE      FALSE
## 6      TRUE      TRUE      TRUE      FALSE      TRUE      FALSE FALSE FALSE      FALSE
##      CLASIZOVE50 PROPBOR AVEDEBT AGRNT_T GBA6RTT GBA4RTT TRRTTOT EFNRLT
## 10      FALSE      FALSE      FALSE      TRUE      TRUE      FALSE      FALSE FALSE
## 9      FALSE      FALSE      FALSE      TRUE      TRUE      FALSE      FALSE FALSE
## 8      FALSE      FALSE      FALSE      TRUE      TRUE      FALSE      FALSE FALSE
## 7      FALSE      FALSE      FALSE      TRUE      TRUE      FALSE      FALSE FALSE
## 6      FALSE      FALSE      FALSE      TRUE      FALSE      FALSE      FALSE FALSE
```

Interpretation of Results

The 6, 7, 8, 9, and 10 variables-models all have high Adjusted R^2 . The top model is:

```
summary(lm(TUITION2 ~ ACTCM50 + STUFACR + ENRTOT + RMINSTTP + RMOUSTTP + SLO6 + CLASIZOVE50 + PROPBOR +
```

```
##
## Call:
## lm(formula = TUITION2 ~ ACTCM50 + STUFACR + ENRTOT + RMINSTTP +
##      RMOUSTTP + SLO6 + CLASIZOVE50 + PROPBOR + AGRNT_T + GBA6RTT,
##      data = df_in)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -20247.0  -7679.3   -338.2   5998.8  26675.8
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.291e+04  2.163e+04   3.833 0.000197 ***
## ACTCM50      8.251e+02  5.259e+02   1.569 0.119119
## STUFACR     -8.158e+02  3.827e+02  -2.132 0.034921 *
## ENRTOT      -4.664e-01  1.232e-01  -3.784 0.000235 ***
## RMINSTTP    -8.361e+02  1.669e+02  -5.009 1.76e-06 ***
## RMOUSTTP    -5.455e+02  1.742e+02  -3.131 0.002154 **
## SLO6        -3.165e+04  1.199e+04  -2.640 0.009325 **
## CLASIZOVE50 -4.955e+04  1.807e+04  -2.742 0.006968 **
## PROPBOR      1.968e+04  7.497e+03   2.625 0.009709 **
## AGRNT_T      2.656e-04  6.088e-05   4.362 2.61e-05 ***
## GBA6RTT      4.000e+02  1.694e+02   2.361 0.019747 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10090 on 129 degrees of freedom
## Multiple R-squared:  0.8616, Adjusted R-squared:  0.8508
## F-statistic: 80.28 on 10 and 129 DF,  p-value: < 2.2e-16
```

```
summary(lm(TUITION3 ~ ACTCM50 + DVADM04 + STUFACR + APPLFEEU + ENRTOT + RMINSTTP + RMOUSTTP + SLO6 + AGRNT_T + GBA6RTT,
```

```
##
## Call:
## lm(formula = TUITION3 ~ ACTCM50 + DVADM04 + STUFACR + APPLFEEU +
##      ENRTOT + RMINSTTP + RMOUSTTP + SLO6 + AGRNT_T + GBA6RTT,
##      data = df_out)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -18086.3  -4451.8   -51.7   3900.1  22826.9
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.054e+04  1.652e+04   3.060  0.002692 **
## ACTCM50      7.454e+02  3.752e+02   1.987  0.049062 *
## DVADM04     -1.162e+02  5.423e+01  -2.143  0.034026 *
## STUFACR     -6.455e+02  2.567e+02  -2.515  0.013144 *
## APPLFEEU     1.371e+02  4.807e+01   2.852  0.005067 **
## ENRTOT      -3.275e-01  7.599e-02  -4.309  3.22e-05 ***
## RMINSTTP    -4.451e+02  1.125e+02  -3.957  0.000125 ***
## RMOUSTTP    -2.171e+02  1.180e+02  -1.839  0.068158 .
## SLO6        -1.398e+04  9.250e+03  -1.512  0.133007
## AGRNT_T      1.826e-04  4.181e-05   4.367  2.56e-05 ***
## GBA6RTT      2.764e+02  1.179e+02   2.344  0.020627 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6971 on 129 degrees of freedom
## Multiple R-squared:  0.8646, Adjusted R-squared:  0.8541
## F-statistic: 82.4 on 10 and 129 DF,  p-value: < 2.2e-16
```

Consistent Drivers (Vertical Columns):

- RMINSTTP (Percent of first-time undergraduates - in-state)
- ENRTOT (Total enrollment)
- AGRNT_T (Total amount of federal, state, local or institutional grant aid awarded to full-time first-time undergraduates)
- STUFACR (Student-faculty ratio)
- GBA6RTT (Graduation rate - Bachelor degree within 6 years, total)

Key Differences:

- In-State: More sensitive to CLASIZOVE50 (Big class size) than Out-of-state
- Out-of-State: More sensitive to ACTCM50 (ACT Composite 50th percentile score) and APPLFEEU (Undergraduate program application fee) than In-state

Scatter Plots: Tuition vs Key Predictors

Below we plot out-of-state tuition (TUITION3) against each of the best predictors from the subset regression.

```
# Long-format for faceting: one row per (school, predictor_name, x_value)
key_vars <- c(
  "ACTCM50" = "ACT composite 50th %ile",
  "STUFACR" = "Student-faculty ratio",
  "GBA6RTT" = "6-year graduation rate",
  "ENRTOT" = "Total enrollment",
  "CLASIZOVE50" = "P(classes >50)",
  "PROPBOR" = "Proportion borrowers"
)

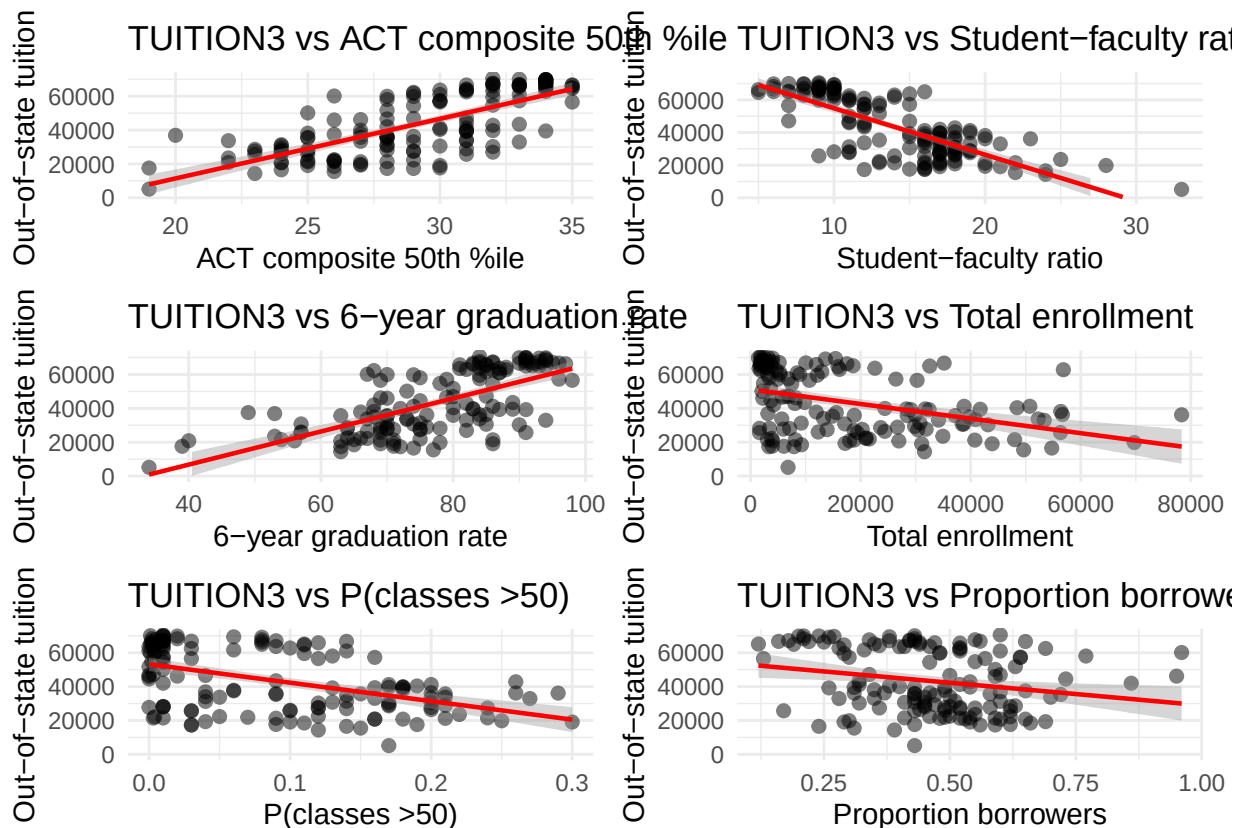
# Out-of-state tuition vs each key predictor (with trend line)
plots_list <- list()
for (i in seq_along(key_vars)) {
  x_var <- names(key_vars)[i]
  x_lab <- unname(key_vars)[i]
  p <- ggplot(df_out, aes(x = .data[[x_var]], y = TUITION3)) +
    geom_point(alpha = 0.5, size = 2) +
    geom_smooth(method = "lm", se = TRUE, color = "red", linewidth = 0.8) +
    scale_y_continuous(limits = c(0, NA)) +
    labs(title = paste0("TUITION3 vs ", x_lab), x = x_lab, y = "Out-of-state tuition") +
    theme_minimal()
  plots_list[[i]] <- p
}

# 2 columns, 3 rows
wrap_plots(plots_list, ncol = 2)

## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 11 rows containing missing values or values outside the scale range
## ('geom_smooth()').

## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'
## 'geom_smooth()' using formula = 'y ~ x'
```

```
# In-state tuition vs same key predictors
plots_list_in <- list()
for (i in seq_along(key_vars)) {
  x_var <- names(key_vars)[i]
  x_lab <- unname(key_vars)[i]
  p <- ggplot(df_in, aes(x = .data[[x_var]], y = TUITION2)) +
    geom_point(alpha = 0.5, size = 2) +
    geom_smooth(method = "lm", se = TRUE, color = "blue", linewidth = 0.8) +
    scale_y_continuous(limits = c(0, NA)) +
    labs(title = paste0("TUITION2 vs ", x_lab), x = x_lab, y = "In-state tuition") +
    theme_minimal()
  plots_list_in[[i]] <- p
}
wrap_plots(plots_list_in, ncol = 2)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 13 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 31 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 20 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```

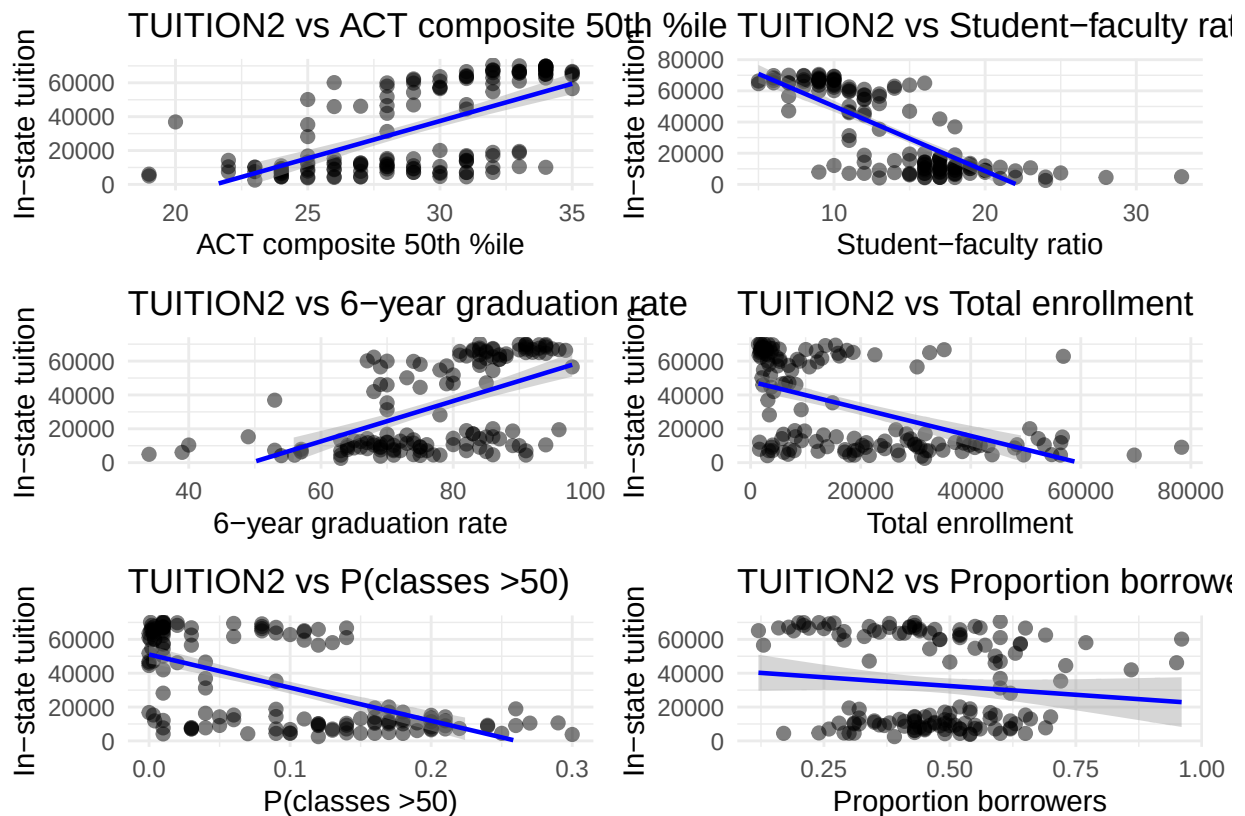
```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 20 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 11 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Holistic Tuition: Overpriced vs Good Value

In-state tuition (TUITION2)

```

# Build analysis data: same sample as prep_data so results match earlier models.
# Keep UNITID and INSTNM so we can label "top overpriced" and "good value" schools in tables.
dat_in <- df %>%
  filter(!is.na(TUITION2), TUITION2 > 0) %>%
  select(UNITID, INSTNM, TUITION2, all_of(VARS_TO_KEEP)) %>%
  mutate(across(any_of(c("SLO6", "SLOA")), ~ ifelse(. == "Yes", 1, 0))) %>%
  select(UNITID, INSTNM, TUITION2, ACTCM50, STUFACR, ENRTOT, RMINSTP, RMOUSTTP, SLO6, CLASIZOVE50, PROPBOR)
na.omit()

# Fit best 10-predictor model; residual = actual - predicted (positive = charges more than expected).
lm_in <- lm(TUITION2 ~ ACTCM50 + STUFACR + ENRTOT + RMINSTP + RMOUSTTP + SLO6 + CLASIZOVE50 + PROPBOR,
  data = select(dat_in, -UNITID, -INSTNM))
dat_in <- dat_in %>%
  mutate(pred = predict(lm_in), resid = TUITION2 - pred) %>%
  mutate(value_tier = case_when(
    resid >= quantile(resid, 0.9) ~ "Overpriced", # top 10% residual = overpriced
    resid <= quantile(resid, 0.1) ~ "Good value", # bottom 10% = good value
    TRUE ~ "Fair value"
  ))

# Points above the 45 degree line: actual > predicted (overpriced). Below: good value.
ggplot(dat_in, aes(x = pred, y = TUITION2)) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray50") +
  geom_point(aes(color = value_tier), alpha = 0.7, size = 2) +
  scale_color_manual(values = c("Overpriced" = "red", "Good value" = "green", "Fair value" = "gray")) +
  scale_x_continuous(limits = c(0, NA)) +
  scale_y_continuous(limits = c(0, NA)) +
  labs(title = "TUITION2: Actual vs predicted (in-state)", x = "Predicted tuition", y = "Actual tuition")
theme_minimal()

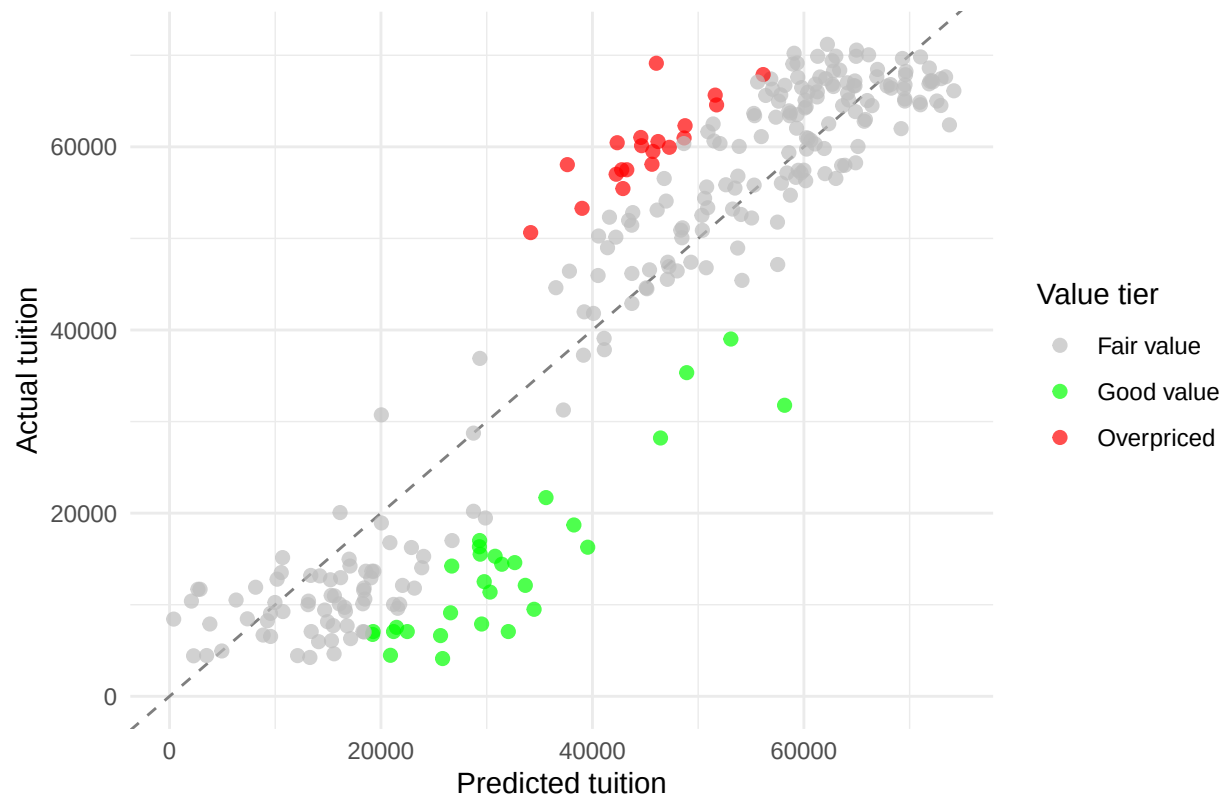
```

```

## Warning: Removed 13 rows containing missing values or values outside the scale range
## ('geom_point()').

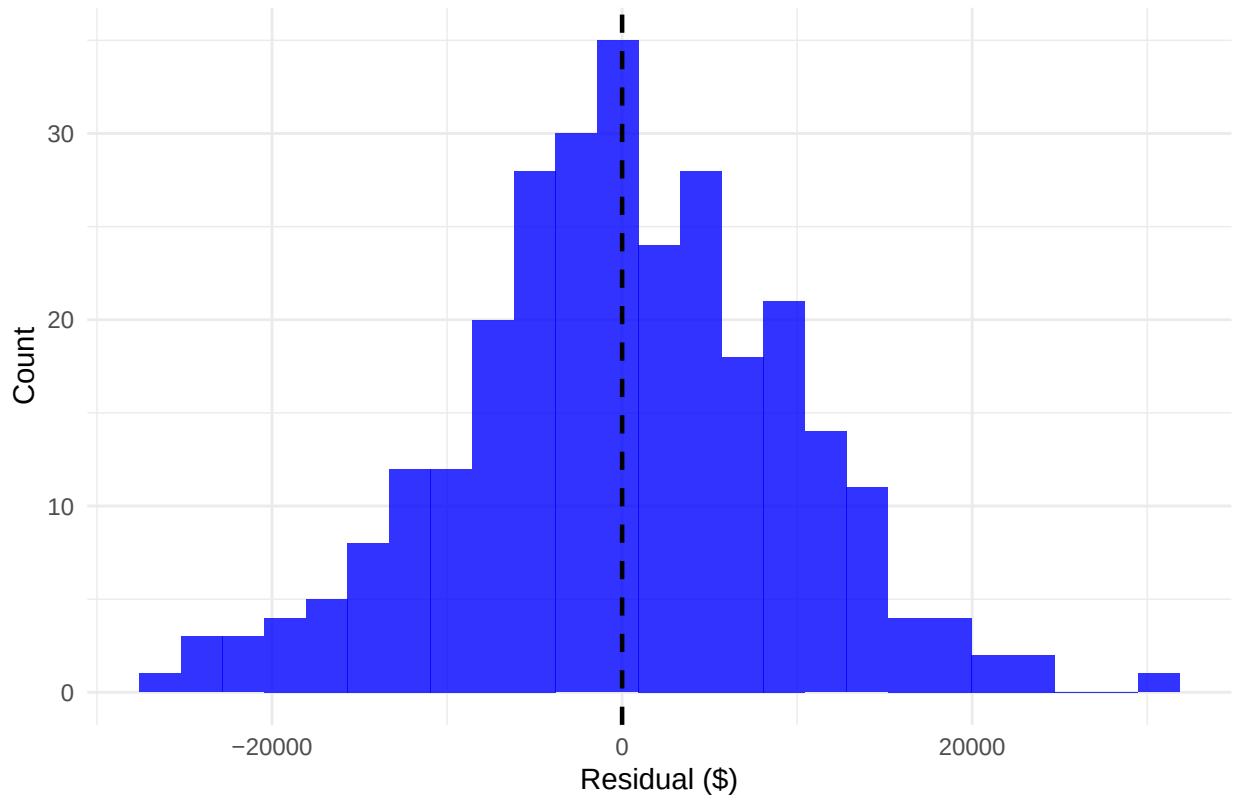
```

TUITION2: Actual vs predicted (in-state)



```
# Residuals centered near 0; right tail = overpriced, left tail = good value.
ggplot(dat_in, aes(x = resid)) +
  geom_histogram(bins = 25, fill = "blue", alpha = 0.8) +
  geom_vline(xintercept = 0, linetype = "dashed", linewidth = 0.8) +
  labs(title = "TUITION2: Residuals (actual - predicted)", x = "Residual ($)", y = "Count") +
  theme_minimal()
```

TUITION2: Residuals (actual – predicted)



```
# Rank by residual: highest = most overpriced, lowest = best value.
top_over_in <- dat_in %>% arrange(desc(resid)) %>% slice_head(n = 10) %>% select(INSTNM, TUITION2, pred, resid)
top_value_in <- dat_in %>% arrange(resid) %>% slice_head(n = 10) %>% select(INSTNM, TUITION2, pred, resid)
knitr::kable(top_over_in, caption = "Top 10 overpriced (TUITION2)", digits = 0)
```

Table 1: Top 10 overpriced (TUITION2)

INSTNM	TUITION2	pred	resid
University of Central Florida	4478	-26172	30650
Florida International University	4721	-18602	23323
Pepperdine University	69130	46045	23085
University of Houston	8685	-12512	21197
Clarkson University	58058	37628	20430
The University of Texas at Dallas	10639	-9186	19825
Texas A & M University-College Station	9092	-9961	19053
University of North Carolina at Charlotte	3812	-14547	18359
Kalamazoo College	60453	42344	18109
Illinois Institute of Technology	50636	34159	16477

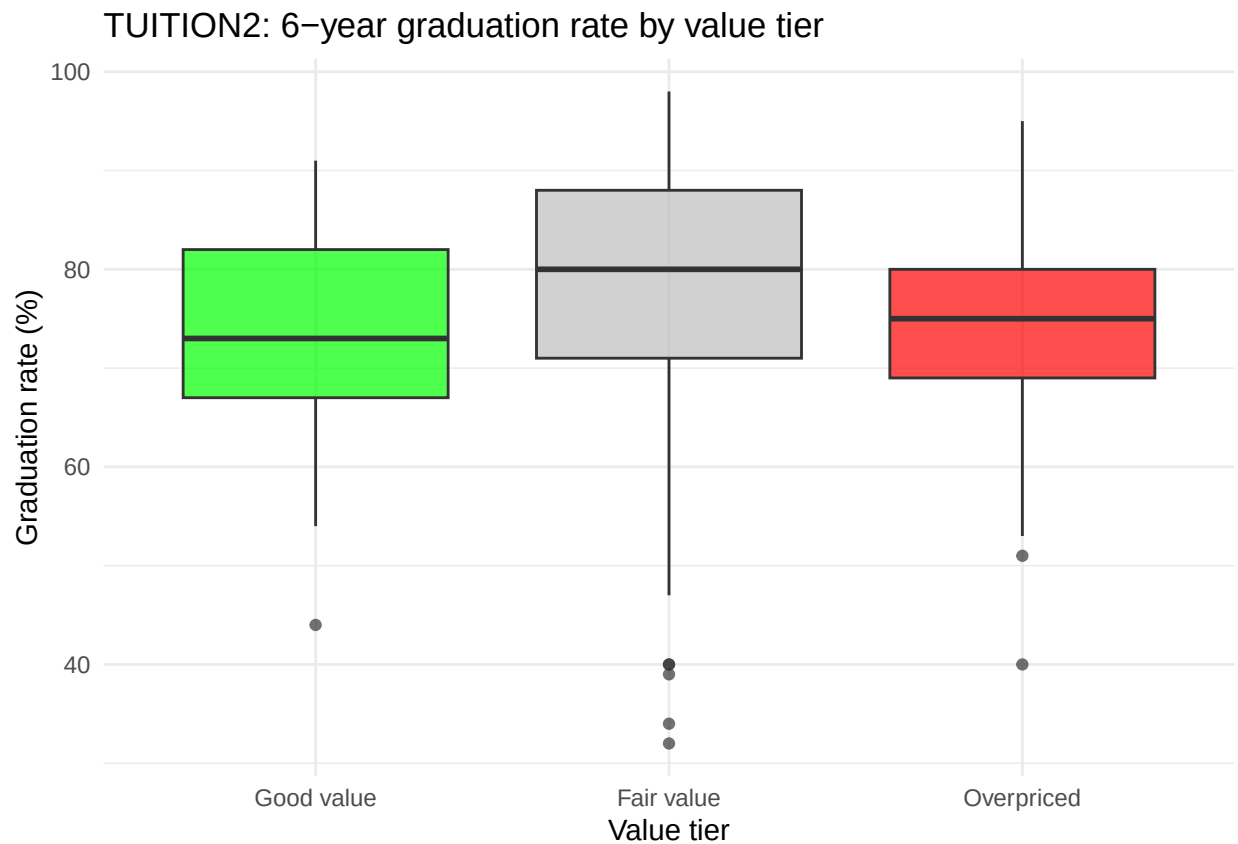
```
knitr::kable(top_value_in, caption = "Top 10 good value (TUITION2)", digits = 0)
```

Table 2: Top 10 good value (TUITION2)

INSTNM	TUITION2	pred	resid
Hillsdale College	31780	58176	-26396
Truman State University	9500	34482	-24982
SUNY at Purchase College	7070	32045	-24975
University of Vermont	16280	39554	-23274
University of North Carolina Asheville	4122	25828	-21706
New Mexico Institute of Mining and Technology	7894	29520	-21626
St. Mary's College of Maryland	12116	33652	-21536
William & Mary	18709	38247	-19538
University of Science and Arts of Oklahoma	6630	25634	-19004
The University of Alabama	11380	30319	-18939

Check whether "overpriced" schools have higher graduation rates (outcome by value tier).

```
ggplot(dat_in, aes(x = value_tier, y = GBA6RTT, fill = value_tier)) +
  geom_boxplot(alpha = 0.7) +
  scale_fill_manual(values = c("Overpriced" = "red", "Good value" = "green", "Fair value" = "gray")) +
  scale_x_discrete(limits = c("Good value", "Fair value", "Overpriced")) +
  labs(title = "TUITION2: 6-year graduation rate by value tier", x = "Value tier", y = "Graduation rate") +
  theme_minimal() + theme(legend.position = "none")
```



Test: does mean graduation rate differ across value tiers? Significant F suggests tiers are not equivalent.

```
anova_in <- aov(GBA6RTT ~ value_tier, data = dat_in)
```

```
summary(anova_in)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## value_tier    2   1268   633.9   4.221 0.0156 *
## Residuals   287  43099   150.2
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Out-of-state tuition (TUITION3)

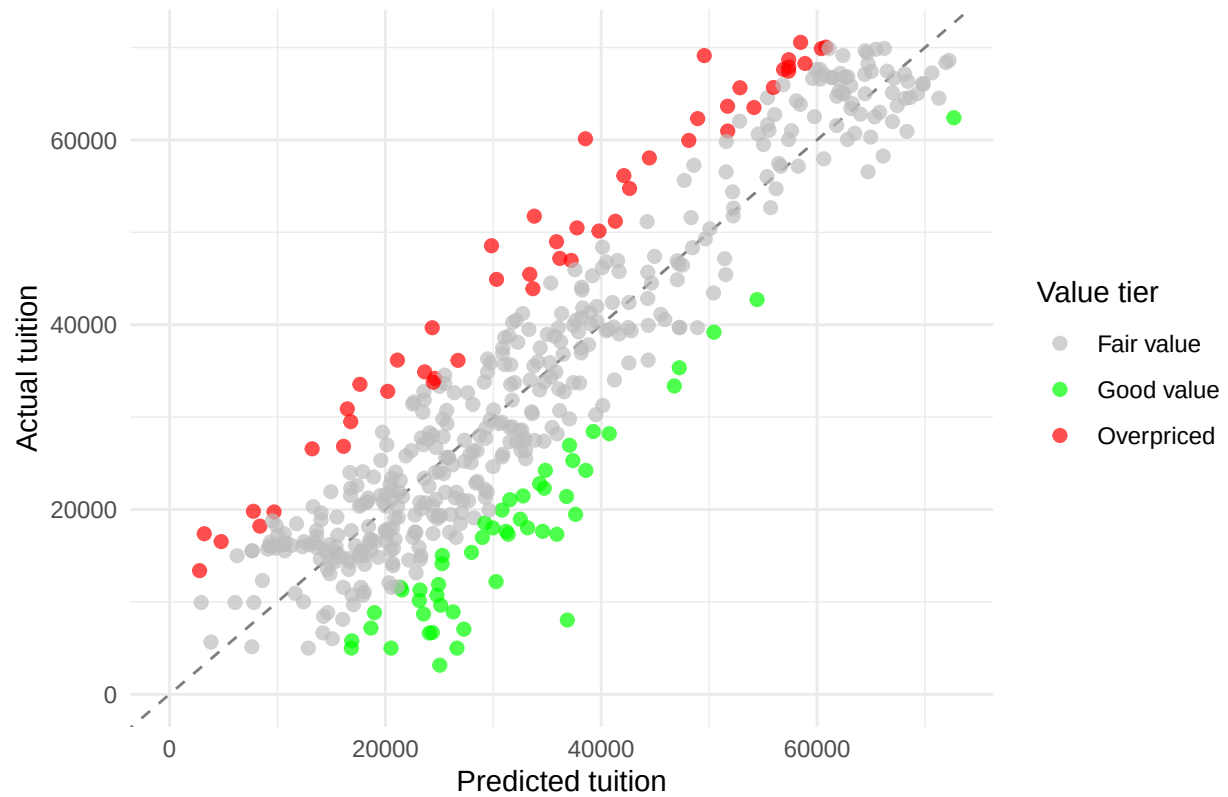
```
# Same logic as TUITION2: same sample definition, keep INSTNM for tables, use best 10-predictor model f
dat_out_holistic <- df %>%
  filter(!is.na(TUITION3), TUITION3 > 0) %>%
  select(UNITID, INSTNM, TUITION3, all_of(VARS_TO_KEEP)) %>%
  mutate(across(any_of(c("SLO6", "SLOA")), ~ ifelse(. == "Yes", 1, 0))) %>%
  select(UNITID, INSTNM, TUITION3, ACTCM50, DVADM04, STUFACR, APPLFEEU, ENRTOT, RMINSTTP, RMOUSTTP, SLO6, SLOA,
  na.omit()
```

```
lm_out <- lm(TUITION3 ~ ACTCM50 + DVADM04 + STUFACR + APPLFEEU + ENRTOT + RMINSTTP + RMOUSTTP + SLO6 + SLOA,
  data = select(dat_out_holistic, -UNITID, -INSTNM))
dat_out_holistic <- dat_out_holistic %>%
  mutate(pred = predict(lm_out), resid = TUITION3 - pred) %>%
  mutate(value_tier = case_when(
    resid >= quantile(resid, 0.9) ~ "Overpriced",
    resid <= quantile(resid, 0.1) ~ "Good value",
    TRUE ~ "Fair value"
  ))
```

```
# Same as TUITION2: 45° line separates overpriced (above) from good value (below).
ggplot(dat_out_holistic, aes(x = pred, y = TUITION3)) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray50") +
  geom_point(aes(color = value_tier), alpha = 0.7, size = 2) +
  scale_color_manual(values = c("Overpriced" = "red", "Good value" = "green", "Fair value" = "gray")) +
  scale_x_continuous(limits = c(0, NA)) +
  scale_y_continuous(limits = c(0, NA)) +
  labs(title = "TUITION3: Actual vs predicted (out-of-state)", x = "Predicted tuition", y = "Actual tuition") +
  theme_minimal()
```

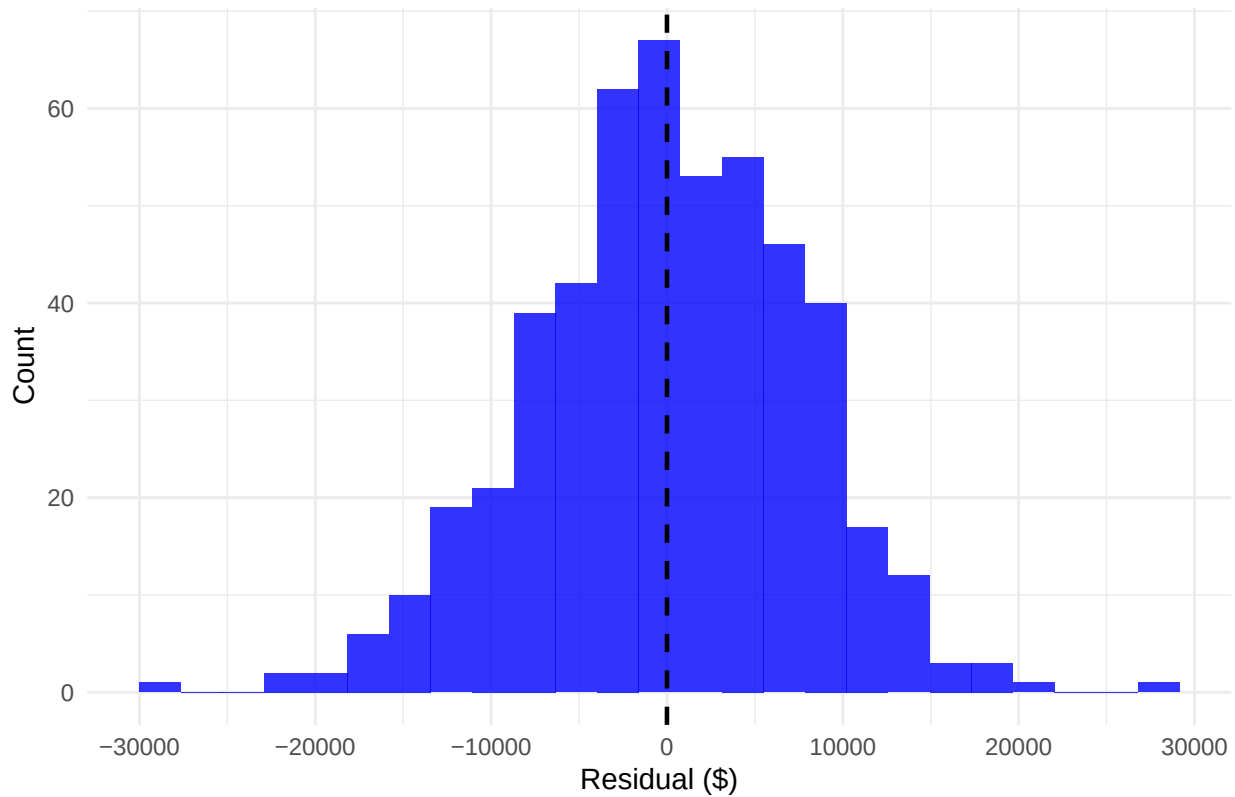
```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```

TUITION3: Actual vs predicted (out-of-state)



```
# Residual distribution; interpretation same as TUITION2.
ggplot(dat_out_holistic, aes(x = resid)) +
  geom_histogram(bins = 25, fill = "blue", alpha = 0.8) +
  geom_vline(xintercept = 0, linetype = "dashed", linewidth = 0.8) +
  labs(title = "TUITION3: Residuals (actual - predicted)", x = "Residual ($)", y = "Count") +
  theme_minimal()
```


TUITION3: Residuals (actual – predicted)



```
# Rank by residual; show top 10 overpriced and top 10 best value.
top_over_out  <- dat_out_holistic %>% arrange(desc(resid)) %>% slice_head(n = 10) %>% select(INSTNM, TUITION3, resid)
top_value_out <- dat_out_holistic %>% arrange(resid) %>% slice_head(n = 10) %>% select(INSTNM, TUITION3, resid)
knitr::kable(top_over_out, caption = "Top 10 overpriced (TUITION3)", digits = 0)
```

Table 3: Top 10 overpriced (TUITION3)

INSTNM	TUITION3	pred	resid
DeVry University-Illinois	16448	-11509	27957
University of Redlands	60128	38539	21589
Pepperdine University	69130	49550	19580
University of La Verne	48550	29835	18715
California Lutheran University	51750	33801	17949
Friends University	33552	17643	15909
University of Connecticut-Hartford Campus	39678	24342	15336
Texas A & M University-College Station	36169	21118	15051
The University of Texas at Austin	44908	30305	14603
Muskingum University	30894	16475	14419

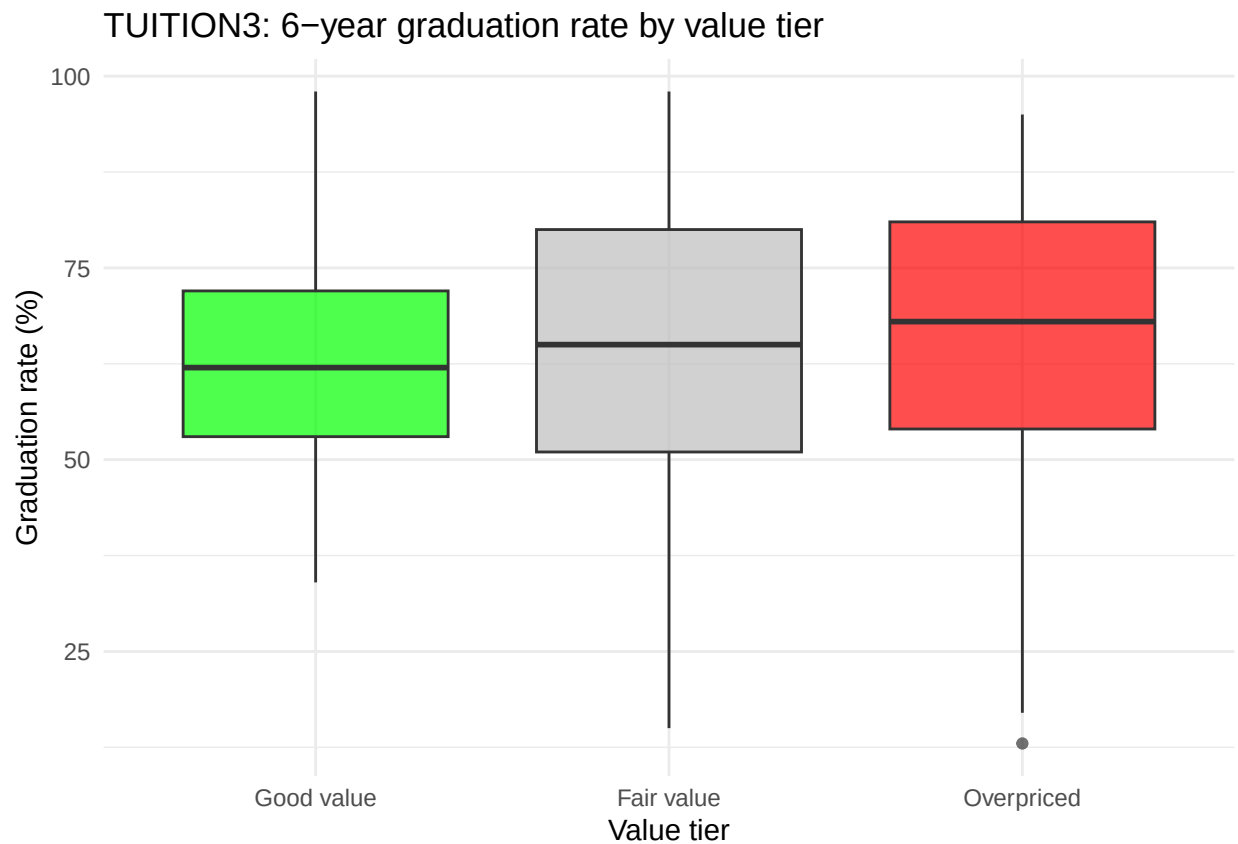
```
knitr::kable(top_value_out, caption = "Top 10 good value (TUITION3)", digits = 0)
```

Table 4: Top 10 good value (TUICTION3)

INSTNM	TUICTION3	pred	resid
Louisiana State University and Agricultural & Mechanical College	8038	36864	-28826
Florida Polytechnic University	3152	25045	-21893
Western Carolina University	5000	26639	-21639
Bridgewater State University	7050	27274	-20224
SUNY at Purchase College	17320	35891	-18571
Touro University	19472	37624	-18152
South Dakota School of Mines and Technology	12200	30260	-18060
Brigham Young University	6688	24347	-17659
Brigham Young University-Hawaii	6630	24070	-17440
Southern Illinois University-Edwardsville	8918	26283	-17365

Outcome (graduation rate) by value tier; same idea as TUICTION2.

```
ggplot(dat_out_holistic, aes(x = value_tier, y = GBA6RTT, fill = value_tier)) +
  geom_boxplot(alpha = 0.7) +
  scale_fill_manual(values = c("Overpriced" = "red", "Good value" = "green", "Fair value" = "gray")) +
  scale_x_discrete(limits = c("Good value", "Fair value", "Overpriced")) +
  labs(title = "TUICTION3: 6-year graduation rate by value tier", x = "Value tier", y = "Graduation rate") +
  theme_minimal() + theme(legend.position = "none")
```



Test whether mean graduation rate differs by value tier for out-of-state tuition.

```
anova_out <- aov(GBA6RTT ~ value_tier, data = dat_out_holistic)
```

```
summary(anova_out)
```

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## value_tier    2    100    50.0   0.147  0.864
## Residuals  499 170051   340.8
```

Summary

Basically what I did is:

1. Predict in-state and out-of-state tuition based on 10 predictors from the best-subset model.
2. Residual = actual - predicted. So positive = overpriced schools, and negative = good value schools because they charge less than expected.
3. I created tiers by saying the top 10% are overpriced and the bottom 10% are underpriced and kept the rest as “fair value”.
4. Added tables that show the names of the top 10 most overpriced and most underpriced schools.